



Naïve Bayes Model

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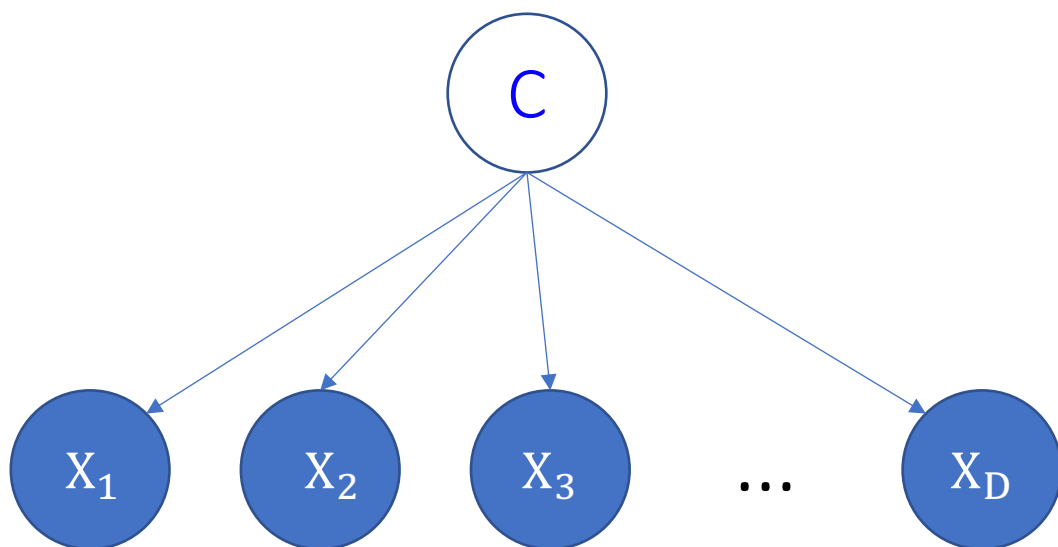


Content

- ❖ Naïve Bayes: what?
- ❖ Naïve Bayes: parameter estimation
- ❖ Naïve Bayes: applications



Naïve Bayes: what?



Naïve Bayes assumption:

$$X_i \perp \mathcal{X}_{-i} \mid C$$

$$\text{Với: } \mathcal{X}_{-i} = \mathcal{X} - \{X_i\}$$

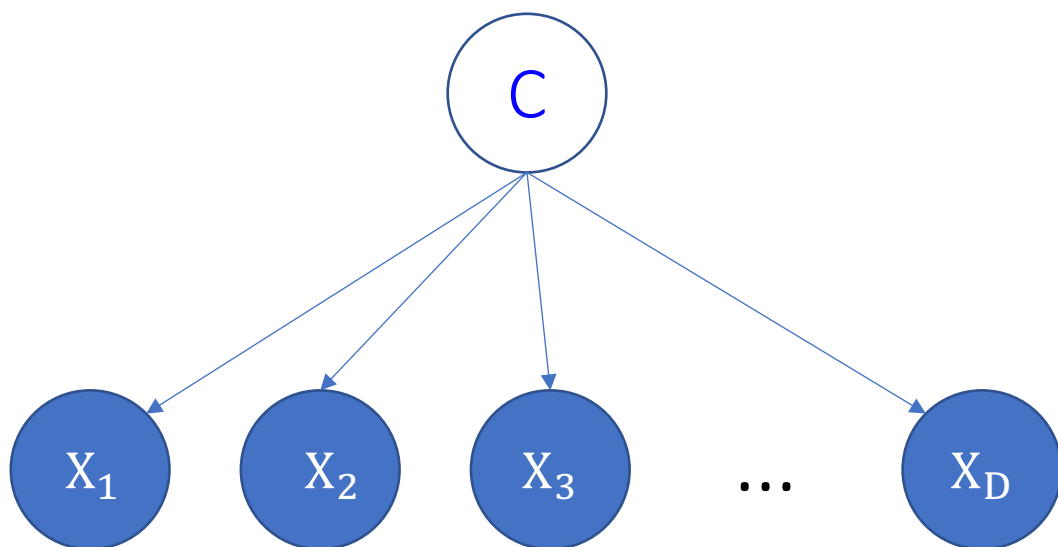
(X_i is independent with remaining variables given C)
(conditional independence)

Naïve Bayes is a simple Bayes Network

$\mathcal{X} = \{X_1, X_2, \dots, X_D\}$: set of variables/features/attributes



Naïve Bayes: what?



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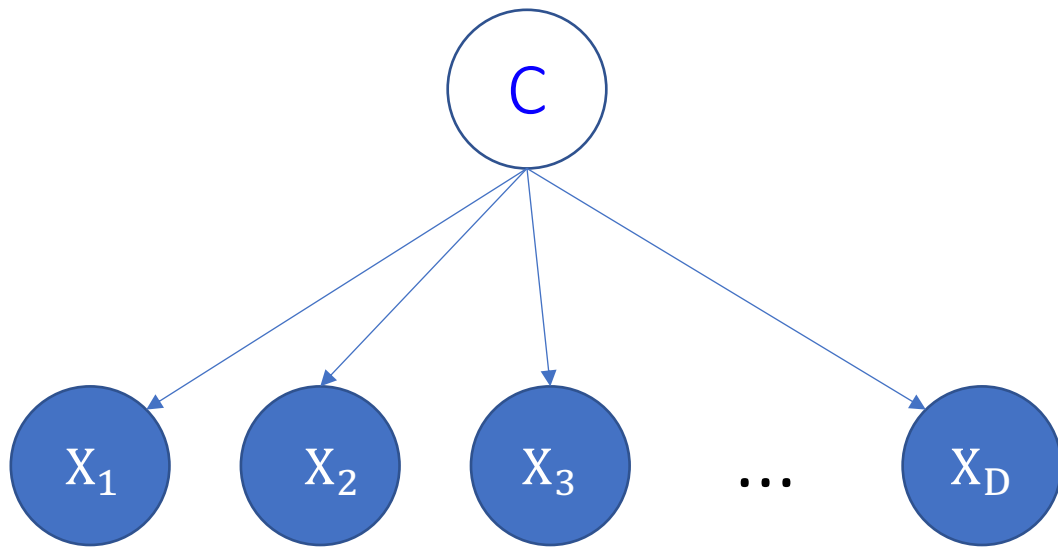
(X_i is independent with remaining variables given C)
(conditional independence)



Reason of “naïve” (ngây thơ)
– too tight assumption



Naïve Bayes: what?



Naïve Bayes is a simple Bayes Network

$\mathcal{X} = \{X_1, X_2, \dots, X_D\}$: set of variables/features/attributes

Naïve Bayes assumption:

$$X_i \perp \mathcal{X}_{-i} \mid C$$

Với: $\mathcal{X}_{-i} = \mathcal{X} - \{X_i\}$

(X_i is independent with remaining variables given C)
(conditional independence)

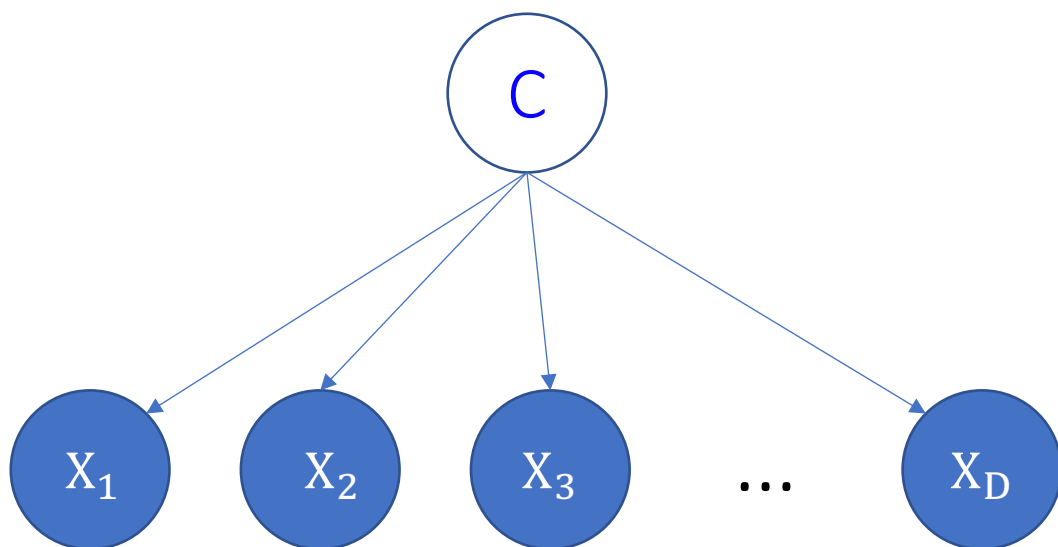


$$P(C, X_1, X_2, \dots, X_D) = P(C) \prod_{i=1}^D P(X_i \mid C)$$



Naïve Bayes: what?

Model's parameters



Model's parameters = set of distributions

- $P(C)$
- $P(X_i | C)$ for $i=1,2,\dots, D$

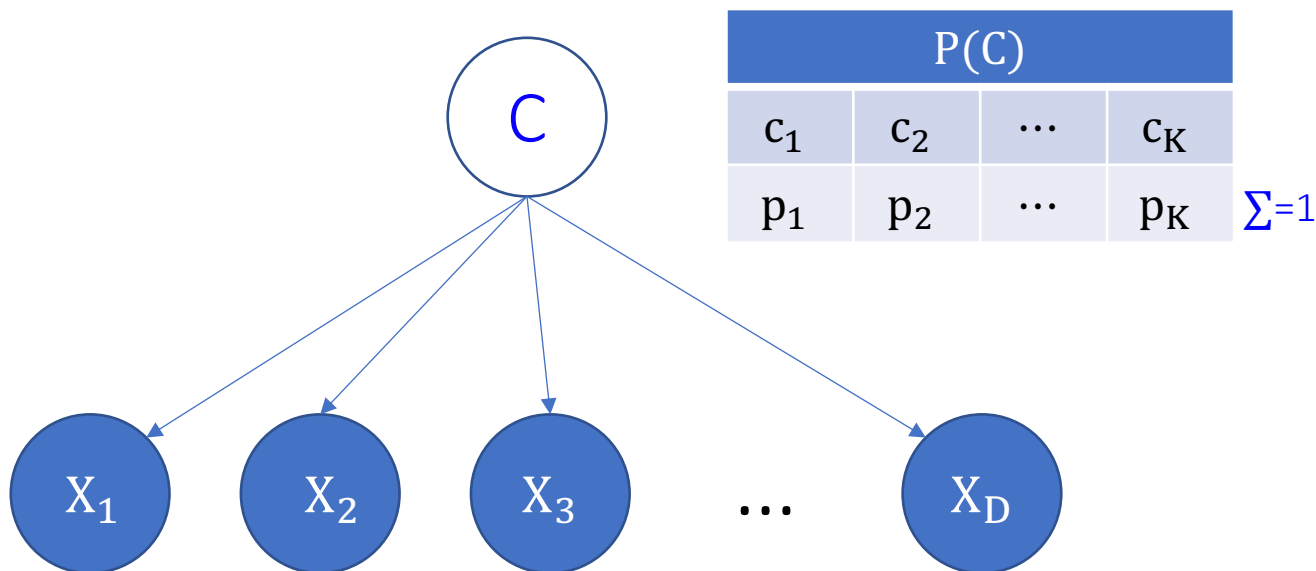
Naïve Bayes is a simple Bayes Network

$\mathcal{X} = \{X_1, X_2, \dots, X_D\}$: set of
variables/features/attributes



Naïve Bayes: what?

Model's parameters: discrete variables, categorical distributions



Naïve Bayes is a simple Bayes Network

Conditional distributions:



P(C)			
c_1	c_2	...	c_K
p_1	p_2	...	p_K

$\Sigma=1$

P(X_1 C)		
C	X_1	P
c_1	$x_{1,1}$	$p_{1,1}$
c_1	$x_{1,2}$	$p_{1,2}$
c_1	...	...
c_1	x_{1,F_1}	p_{1,F_1}
c_2	$x_{2,1}$	$p_{2,1}$
c_2	$x_{2,2}$	$p_{2,2}$
c_2	...	...
c_2	x_{2,F_2}	p_{2,F_2}
...	...	...

$\Sigma=1$

...

P(X_D C)		
C	X_D	P
c_1	$x_{D,1}$	$p_{D,1}$
c_1	$x_{D,2}$	$p_{D,2}$
c_1	...	...
c_1	x_{D,F_D}	p_{D,F_D}
c_2	$x_{D,1}$	$p_{D,1}$
c_2	$x_{D,2}$	$p_{D,2}$
c_2	...	...
c_2	x_{D,F_D}	p_{D,F_D}
...	...	...

$\Sigma=1$

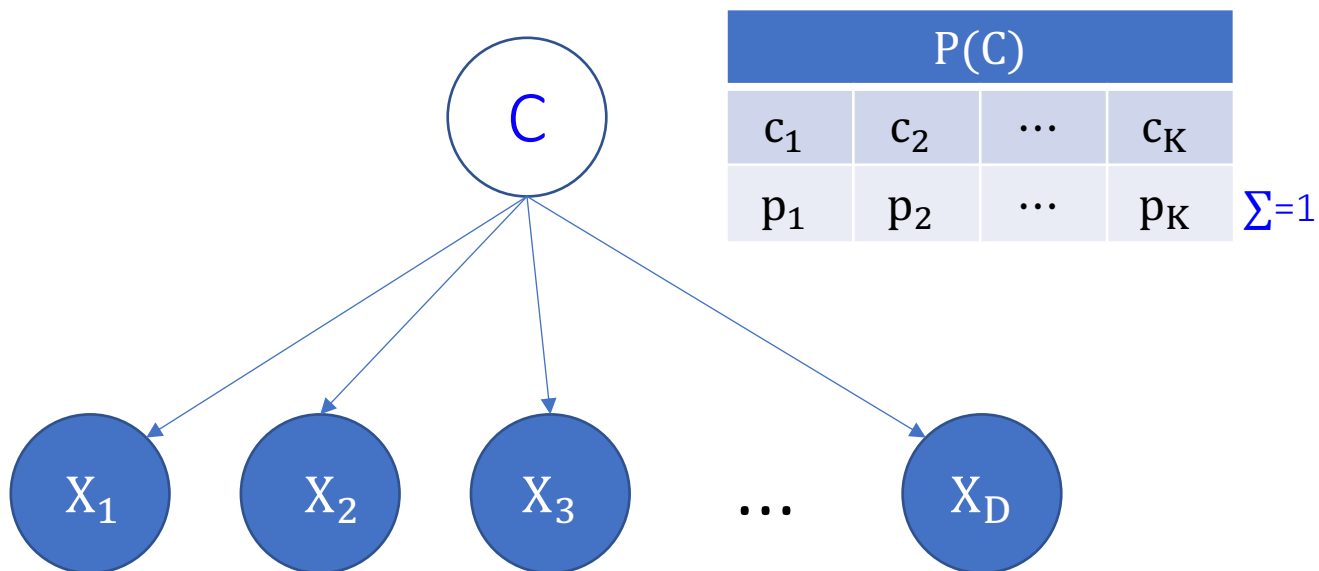
- $K = |\text{dom}(C)|$ (num. of classes in classification task)
- D : num of variables (num of features/attributes)
- $F_i = |\text{dom}(X_i)|$



Naïve Bayes: what?

Model's parameters:

discrete variables, multinomial distributions



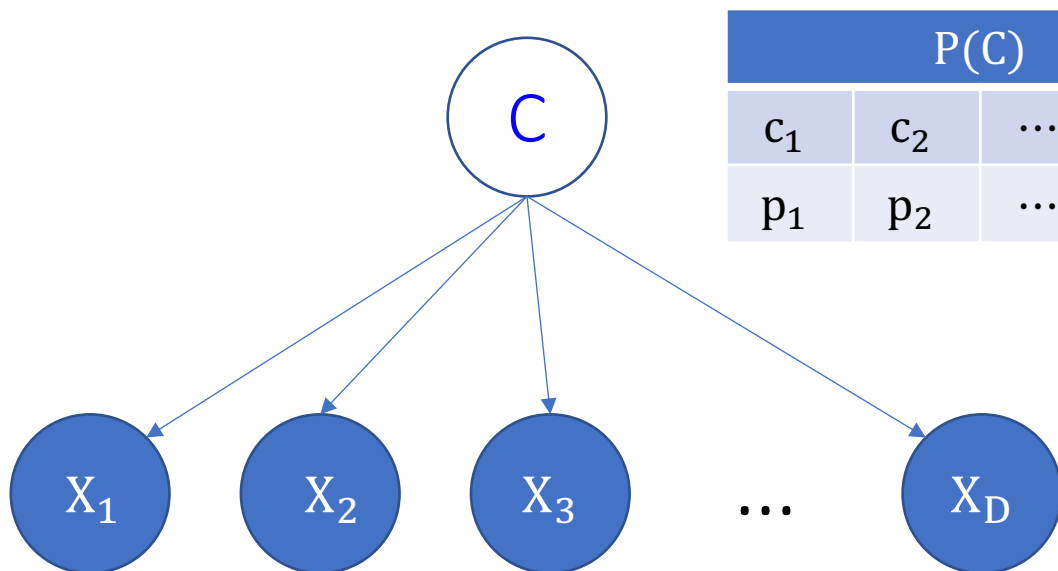
P(X C)					
C	X ₁	X ₂	...	X _D	
c ₁	p _{1,1}	p _{1,2}	...	p _{1,D}	Σ=1 (multinomial)
c ₂	p _{2,1}	p _{2,2}	...	p _{2,D}	Σ=1 (multinomial)
...	...	...	...	...	
c _K	p _{K,1}	p _{K,2}	...	p _{K,D}	Σ=1 (multinomial)

Naïve Bayes is a simple Bayes Network



Naïve Bayes: what?

Model's parameters: continuous variables



Naïve Bayes is a simple Bayes Network

P(C)			
c_1	c_2	...	c_K
p_1	p_2	...	p_K

$\Sigma=1$

P(X_1 C)		
C	X_1	P
c_1	$\mathcal{P}(\theta_{1,1})$	
c_2	$\mathcal{P}(\theta_{2,1})$	
...	...	...
c_K	$\mathcal{P}(\theta_{K,1})$	

...

P(X_D C)		
C	X_D	P
c_1	$\mathcal{P}(\theta_{1,D})$	
c_2	$\mathcal{P}(\theta_{2,D})$	
...	...	...
c_K	$\mathcal{P}(\theta_{K,D})$	

$\mathcal{P}(\theta_{c,d})$: conditional distribution of X_d given $C=c$

- $\theta_{c,d}$: set of parameters; example: (μ, σ) if Gaussian

If $X_1, X_2, \dots, X_D$ are Gaussian $\Rightarrow X = [X_1, X_2, \dots, X_D]$

$P(\mathbf{X} | C)$: multi-variables Gaussian (see sklearn)



Naïve Bayes: what?

For classification problems

Training:

X_1	X_2	$\dots$	X_D	C
$x_1^{(1)}$	$x_2^{(1)}$	$\dots$	$x_D^{(1)}$	$c^{(1)}$
$x_1^{(2)}$	$x_2^{(2)}$	$\dots$	$x_D^{(2)}$	$c^{(2)}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$x_1^{(N)}$	$x_2^{(N)}$	$\dots$	$x_D^{(N)}$	$c^{(N)}$

$$c^{(i)} \in \{c_1, c_2, \dots, c_K\}$$



Model's parameters

Prediction:

X_1	X_2	$\dots$	X_D	C
x_1	x_2	$\dots$	x_D	?



Predict the class c for input:

$$\mathbf{x} = [x_1, x_2, \dots, x_D]$$



Naïve Bayes: what?

For classification problems

Prediction:

X_1	X_2	...	X_D	C
x_1	x_2	...	x_D	?



predict:

$$\mathbf{x} = [x_1, x_2, \dots, x_D]$$

||

1. Compute: $P(C | \mathbf{x})$
2. Determine: $c = \operatorname{argmax}_C P(C | \mathbf{x})$



Naïve Bayes: what?

For classification problems

Bayes's formula:

$$P(C | \mathbf{x}) = \frac{P(C) \times P(\mathbf{x} | C)}{P(\mathbf{x})}$$

Prior probability
(xác suất tiên nghiệm)

Likelihood
(Hệ số khả dĩ/hợp lý)

Posterior probability
(xác suất hậu nghiệm)

Evidence probability
(Xác suất bằng chứng)



Naïve Bayes: what?

For classification problems

Bayes's formula:

$$P(C | \mathbf{x}) = \frac{P(C) \times \prod_{i=1}^D P(x_i | C)}{P(\mathbf{x})}$$



$P(\mathbf{x})$: not dependent on C

$$P(C | \mathbf{x}) \propto \left[P(C) \times \prod_{i=1}^D P(x_i | C) \right]$$



Determine the class for $\mathbf{x}$:

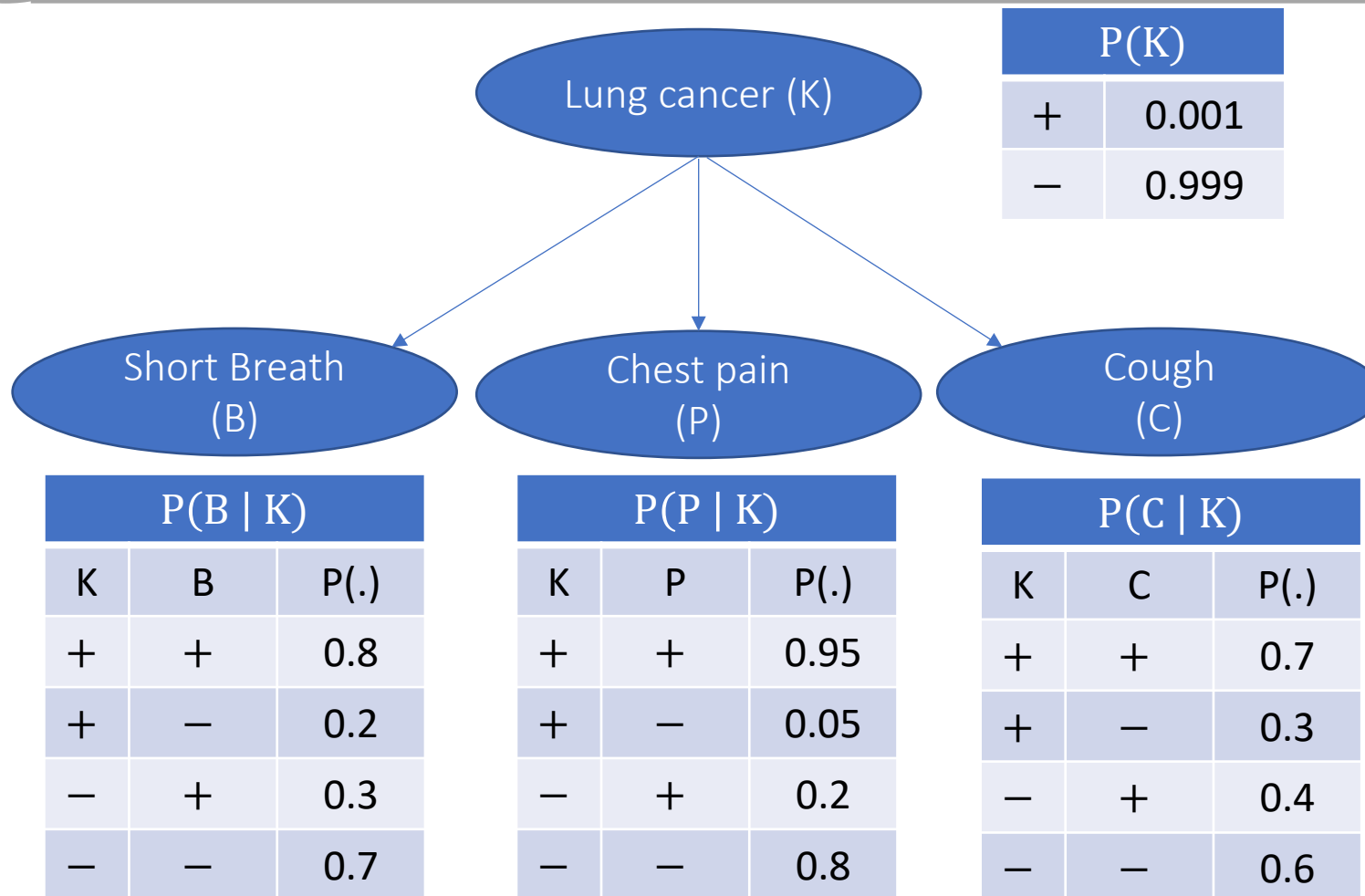
$$c = \operatorname{argmax}_C P(C | \mathbf{x}) = \operatorname{argmax}_C \left[P(C) \times \prod_{i=1}^D P(x_i | C) \right]$$

$$c = \operatorname{argmax}_C \left[\log_2 P(C) + \sum_{i=1}^D \log_2 P(x_i | C) \right]$$



Naïve Bayes: what?

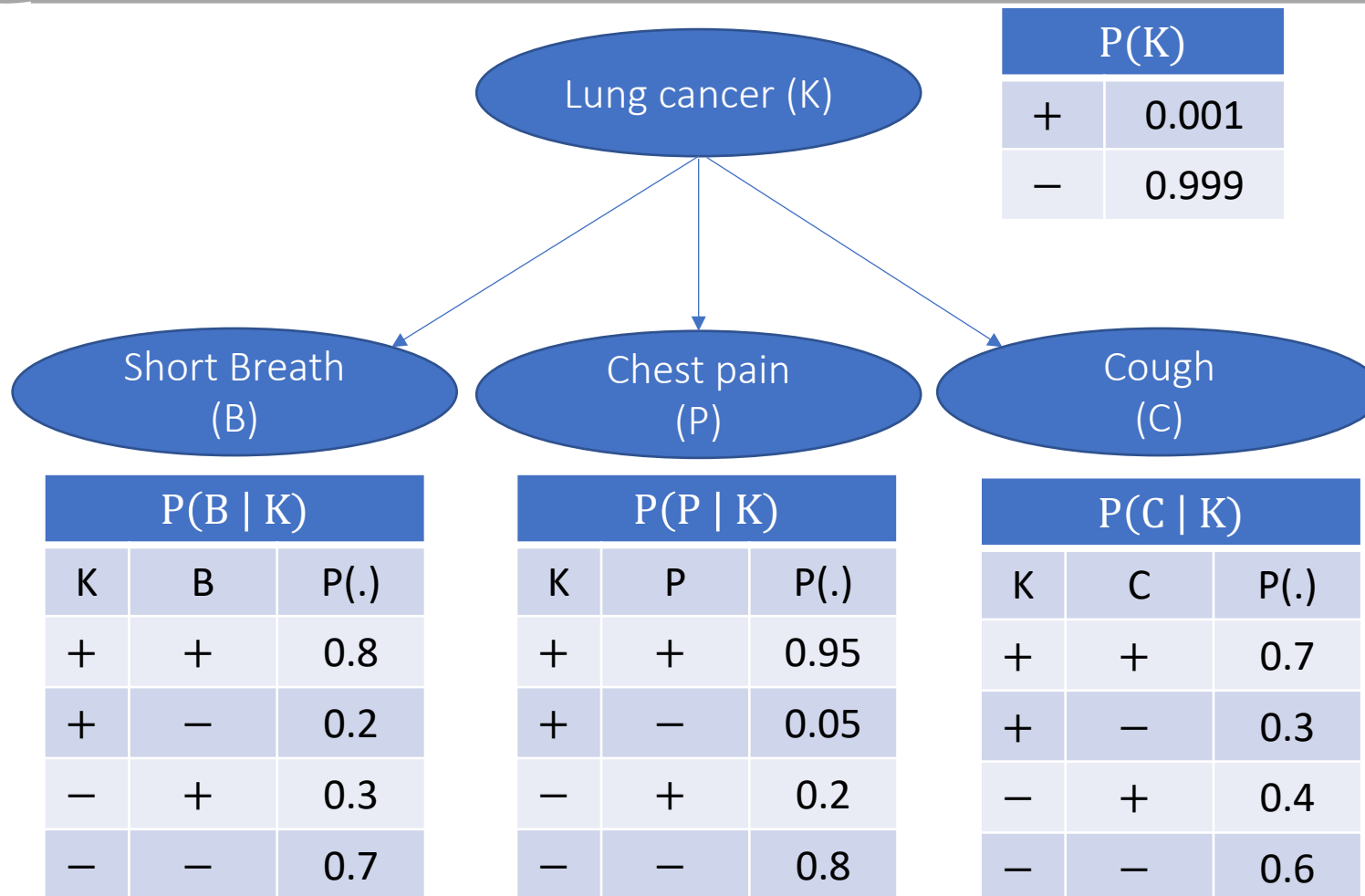
For classification problems: example 1





Naïve Bayes: what?

For classification problems: example 1



Symptoms:

- Short Breath(B): +
- Chest pain (P): +
- Cough (C) : -
- => Features: $\mathbf{x} = [+ , + , -]$

Question: Cancer?



Naïve Bayes: what?

For classification problems: example 1

P(K)	
+	0.001
-	0.999

P(P K)		
K	P	P(.)
+	+	0.95
+	-	0.05
-	+	0.2
-	-	0.8

P(B K)			P(C K)		
K	B	P(.)	K	C	P(.)
+	+	0.8	+	+	0.7
+	-	0.2	+	-	0.3
-	+	0.3	-	+	0.4
-	-	0.7	-	-	0.6

Symptoms:

- Short Breath(B): +
- Chest pain (P): +
- Cough (C) : -
- => Features: $\mathbf{x} = [+ , + , -]$

Question: Cancer?

$$\begin{aligned}P(K = +|\mathbf{x}) &\propto P(C = +) \times P(B = +|+) \times P(P = +|+) \times P(C = -|+) \\&= 0.001 \times 0.8 \times 0.95 \times 0.3 = 0.000228\end{aligned}$$

$$\begin{aligned}P(K = -|\mathbf{x}) &\propto P(C = -) \times P(B = +|-) \times P(P = +|-) \times P(C = -|-) \\&= 0.999 \times 0.3 \times 0.2 \times 0.6 = 0.035964\end{aligned}$$

$P(K = +|\mathbf{x}) < P(K = -|\mathbf{x})$ ➡ No Cancer!



Naïve Bayes: what?

For classification problems: example 2

P(K)	
+	0.001
-	0.999

Lung Cancer(K)

Short Breath
(B)

P(B K)		
K	B	P(.)
+	+	0.8
+	-	0.2
-	+	0.3
-	-	0.7

Chest Pain
(P)

P(P K)		
K	P	P(.)
+	+	0.95
+	-	0.05
-	+	0.2
-	-	0.8

Cough
(C)

P(C K)		
K	C	P(.)
+	+	0.7
+	-	0.3
-	+	0.4
-	-	0.6

X-ray
(X)

P(X K)		
K	C	P(.)
+	+	0.999
+	-	0.001
-	+	0.002
-	-	0.998

Symptoms: Short Breath(B): +; Chest pain (P): +;
Cough (C) : -; X-Ray (X): +

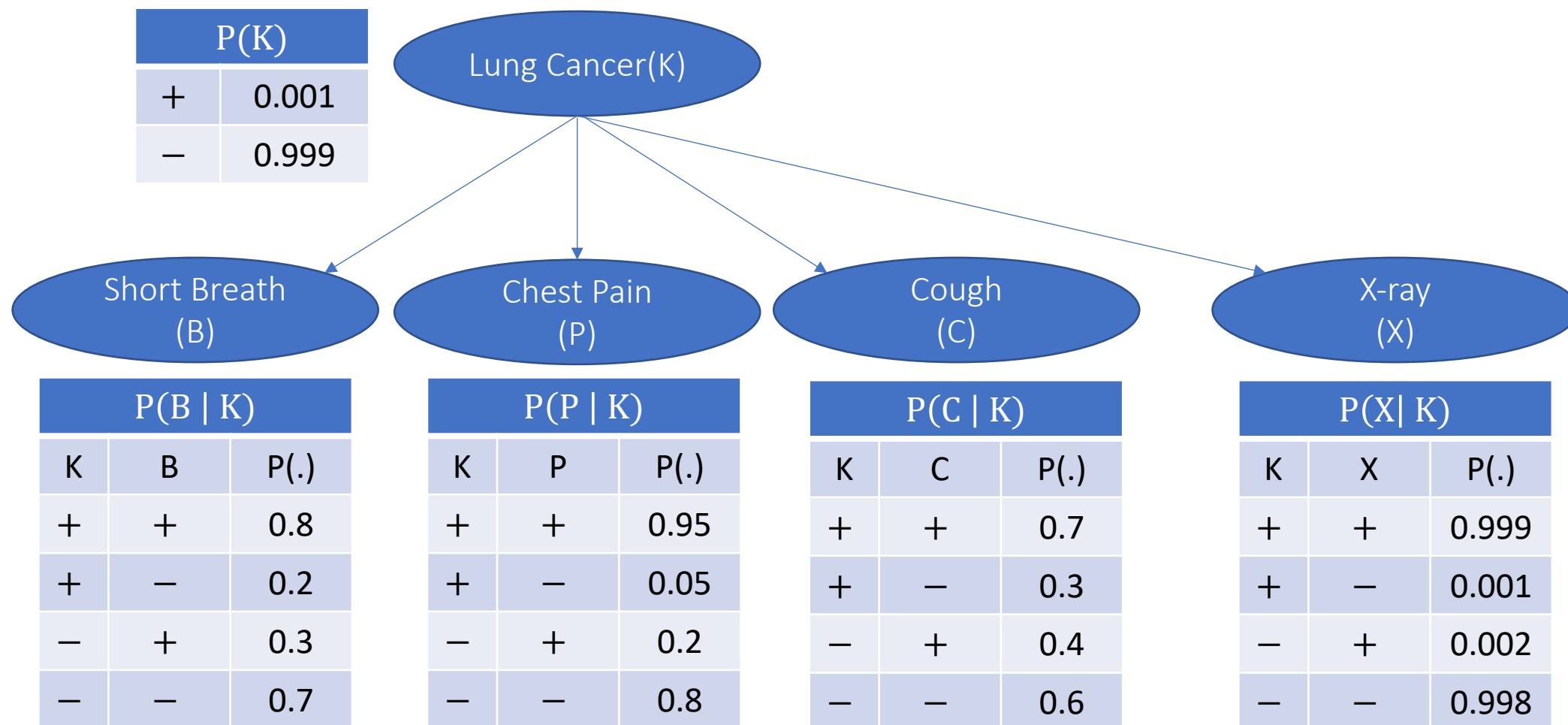
• => Features: $x = [+ , + , - , +]$

Question: Cancer?



Naïve Bayes: what?

For classification problems: example 2





Naïve Bayes: what?

For classification problems: example 2

P(K)	
+	0.001
-	0.999

P(P K)		
K	P	P(.)
+	+	0.95
+	-	0.05
-	+	0.2
-	-	0.8

P(X K)		
K	C	P(.)
+	+	0.999
+	-	0.001
-	+	0.002
-	-	0.998

Symptoms: Short Breath(B): +; Chest pain (P): +;
Cough (C) : -; X-Quang (X): +

• => Features: $\mathbf{x} = [+ , + , - , +]$

Question: Cancer?

P(B K)		
K	B	P(.)
+	+	0.8
+	-	0.2
-	+	0.3
-	-	0.7

P(C K)		
K	C	P(.)
+	+	0.7
+	-	0.3
-	+	0.4
-	-	0.6

$$\begin{aligned} P(K = +|\mathbf{x}) &\propto P(C = +) \times P(B = +|+) \times P(P = +|+) \times \\ &\quad P(C = -|+) \times P(X = +|+) \\ &= 0.001 \times 0.8 \times 0.95 \times 0.3 \times 0.999 = 0.00023 \end{aligned}$$

$$\begin{aligned} P(K = -|\mathbf{x}) &\propto P(C = -) \times P(B = +|-) \times P(P = +|-) \times \\ &\quad P(C = -|-) \times P(X = +|-) \\ &= 0.999 \times 0.3 \times 0.2 \times 0.6 \times 0.002 = 0.000072 \end{aligned}$$

Parameter estimation

$P(X_i|C)$: Categorical distribution



Naïve Bayes: parameter estimation

Categorical distribution

Features :

Outlook (O): {Sunny, Overcast, Rain}

Temperature(T): {Hot, Mild, Cool}

Humidity(H): {High, Normal}

Wind(W): {Strong, Weak}

Target:

Play(C): {Yes (+), No(-)}

STT	Outlook	Temperature	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No



Naïve Bayes: parameter estimation

Categorical distribution

STT	Outlook	Temperature	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

- $N = 14$ (num of rows)
- Histogram of "Play": {Yes: 9, No: 5}
- $P(\text{Play})$: Categorical Distribution
- $P(\text{Play}) = \{\text{Yes: } 9/14, \text{No: } 5/14\}$



Naïve Bayes: parameter estimation

Categorical distribution

STT	Outlook	Temperature	Humidity	Wind	Play
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
7	Overcast	Cool	Normal	Strong	Yes
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes

← Sub-Dataset: given Play=Yes
(9 rows)

Sub-Dataset: given Play=No
(5 rows)

STT	Outlook	Temperature	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
6	Rain	Cool	Normal	Strong	No
8	Sunny	Mild	High	Weak	No
14	Rain	Mild	High	Strong	No



Naïve Bayes: parameter estimation

Categorical distribution

STT	Outlook	Temperature	Humidity	Wind	Play
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
7	Overcast	Cool	Normal	Strong	Yes
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes

Sub-Dataset: given Play=Yes
(9 rows)

- $N_y = 9$
- Histogram of "Outlook" | Play = Yes:
{Overcast: 4, Rain:3, Sunny: 2}
- Histogram of "Outlook" | Play = Yes
(smoothed with $\alpha = 1$):
{Overcast: 4+1, Rain:3+1, Sunny: 2+1}
- N_y (smoothed) = 12
- $P(\text{Outlook} | \text{Play})$: Categorical Distribution
- $P(\text{Outlook} | \text{Play=Yes})$:
{Overcast: 5/12, Rain:4/12, Sunny: 3/12}



Naïve Bayes: parameter estimation

Categorical distribution

STT	Outlook	Temperature	Humidity	Wind	Play
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
7	Overcast	Cool	Normal	Strong	Yes
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes

Sub-Dataset: given Play=Yes
(9 rows)

- $N_y = 9$
- Histogram of "Temperature" | Play = Yes:
{Hot: 2, Mild: 4, Cool: 3}
- Histogram of "Temperature" | Play = Yes
(smoothed with $\alpha = 1$):
{Hot: 2+1, Mild: 4+1, Cool: 3+1}
- N_y (smoothed) = 12
- $P(\text{Temperature} | \text{Play})$: Categorical Distribution
- $P(\text{Temperature} | \text{Play=Yes})$:
{Hot: 3/12, Mild: 5/12, Cool: 4/12}



Naïve Bayes: parameter estimation

Categorical distribution

STT	Outlook	Temperature	Humidity	Wind	Play
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
7	Overcast	Cool	Normal	Strong	Yes
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes

Sub-Dataset: given Play=Yes
(9 rows)

- $N_y = 9$
- Histogram of "Humidity" | Play = Yes:
{High: 3, Normal: 6}
- Histogram of "Humidity" | Play = Yes
(smoothed with $\alpha = 1$):
{High: 3+1, Normal: 6+1}
- N_y (smoothed) = 11
- $P(\text{Humidity} | \text{Play})$: Categorical Distribution
- $P(\text{Humidity} | \text{Play=Yes})$:
{High: 4/11, Normal: 7/11}



Naïve Bayes: parameter estimation

Categorical distribution

STT	Outlook	Temperature	Humidity	Wind	Play
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
7	Overcast	Cool	Normal	Strong	Yes
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes

Sub-Dataset: given Play=Yes
(9 rows)

- $N_y = 9$
- Histogram of "Wind" | Play = Yes:
{Weak: 6, Strong: 3}
- Histogram of "Wind" | Play = Yes
(smoothed with $\alpha = 1$):
{Weak: 6+1, Strong: 3+1}
- N_y (smoothed) = 11
- $P(\text{Wind} | \text{Play})$: Categorical Distribution
- $P(\text{Wind} | \text{Play=Yes})$:
{High: 7/11, Normal: 4/11}



Naïve Bayes: parameter estimation

Categorical distribution

STT	Outlook	Temperature	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
6	Rain	Cool	Normal	Strong	No
8	Sunny	Mild	High	Weak	No
14	Rain	Mild	High	Strong	No

Sub-Dataset: given Play=No
(5 rows)

- $N_y = 5$
- Histogram of "Outlook" | Play = No:
{Overcast: 0, Rain:2, Sunny: 3}
- Histogram of "Outlook" | Play = No
(smoothed with $\alpha = 1$):
{Overcast: 0+1, Rain:2+1, Sunny: 3+1}
- N_y (smoothed) = 8
- $P(\text{Outlook} | \text{Play})$: Categorical Distribution
- $P(\text{Outlook} | \text{Play}=\text{No})$:
{Overcast: 1/8, Rain:3/8, Sunny: 4/8}



Naïve Bayes: parameter estimation

Categorical distribution

STT	Outlook	Temperature	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
6	Rain	Cool	Normal	Strong	No
8	Sunny	Mild	High	Weak	No
14	Rain	Mild	High	Strong	No

Sub-Dataset: given Play=No
(5 rows)

- $N_y = 5$
- Histogram of "Temperature" | Play = No:
{Hot: 2, Cool:1, Mild: 2}
- Histogram of "Temperature" | Play = No
(smoothed with $\alpha = 1$):
{Hot: 2+1, Cool:1+1, Mild: 2+1}
- N_y (smoothed) = 8
- $P(\text{Temperature} | \text{Play})$: Categorical Distribution
- $P(\text{Temperature} | \text{Play=No})$:
{Hot: 3/8, Cool:2/8, Mild: 3/8}



Naïve Bayes: parameter estimation

Categorical distribution

STT	Outlook	Temperature	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
6	Rain	Cool	Normal	Strong	No
8	Sunny	Mild	High	Weak	No
14	Rain	Mild	High	Strong	No

Sub-Dataset: given Play=No
(5 rows)

- $N_y = 5$
- Histogram of "Humidity" | Play = No:
{High: 4, Normal: 1}
- Histogram of "Humidity" | Play = No
(smoothed with $\alpha = 1$):
{High: 4+1, Normal: 1+1}
- N_y (smoothed) = 7
- $P(\text{Humidity} | \text{Play})$: Categorical Distribution
- $P(\text{Humidity} | \text{Play=No})$:
{High: 5/7, Normal: 2/7}



Naïve Bayes: parameter estimation

Categorical distribution

STT	Outlook	Temperature	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
6	Rain	Cool	Normal	Strong	No
8	Sunny	Mild	High	Weak	No
14	Rain	Mild	High	Strong	No

Sub-Dataset: given Play=No
(5 rows)

- $N_y = 5$
- Histogram of "Wind" | Play = No:
{Weak: 2, Strong: 3}
- Histogram of "Wind" | Play = No
(smoothed with $\alpha = 1$):
{Weak: 2+1, Strong: 3+1}
- N_y (smoothed) = 7
- $P(\text{Wind} | \text{Play})$: Categorical Distribution
- $P(\text{Wind} | \text{Play=No})$:
{Weak: 3/7, Strong: 4/7}



Naïve Bayes: parameter estimation

Categorical distribution

Summary:

- $P(\text{Play}) = \{\text{Yes: } 9/14, \text{No: } 5/14\}$
- **Given Play=Yes:**
 - $P(\text{Outlook} | \text{Play=Yes})$:
 $\{\text{Overcast: } 5/12, \text{Rain: } 4/12, \text{Sunny: } 3/12\}$
 - $P(\text{Temperature} | \text{Play=Yes})$:
 $\{\text{Hot: } 3/12, \text{Mild: } 5/12, \text{Cool: } 4/12\}$
 - $P(\text{Humidity} | \text{Play=Yes})$:
 $\{\text{High: } 4/11, \text{Normal: } 7/11\}$
 - $P(\text{Wind} | \text{Play=Yes})$:
 $\{\text{High: } 7/11, \text{Normal: } 4/11\}$
- **Given Play=No:**
 - $P(\text{Outlook} | \text{Play})$:
 $\{\text{Overcast: } 1/8, \text{Rain: } 3/8, \text{Sunny: } 4/8\}$
 - $P(\text{Temperature} | \text{Play=No})$:
 $\{\text{Hot: } 3/8, \text{Cool: } 2/8, \text{Mild: } 3/8\}$
 - $P(\text{Humidity} | \text{Play=No})$:
 $\{\text{High: } 5/7, \text{Normal: } 2/7\}$
 - $P(\text{Wind} | \text{Play=No})$:
 $\{\text{Weak: } 3/7, \text{Strong: } 4/7\}$

Parameter estimation

$P(X_i|C)$: Multinomial Distribution



Naïve Bayes: parameter estimation

Multinomial Distribution, example

Assumption :

- Have $N=12$ documents
- Vocabulary: {word1, word2, word3}
- Classes=Sentiment: {+, -}
- Dataset: count words in each document

IDX	Word1	Word2	Word3	Sentiment
1	4	2	0	+
2	7	3	1	+
3	6	4	3	+
4	5	2	2	+
5	8	5	3	+
6	4	7	5	+
7	9	7	2	+
8	12	2	3	+
9	3	3	7	-
10	0	4	8	-
11	4	3	15	-
12	1	4	10	-



Naïve Bayes: parameter estimation

Multinomial Distribution, example

Sentiment ~ Categorical Distribution

- $N = 12$ (number of documents/rows)
- Histogram of Sentiment: $\{+: 8, -: 4\}$
- $P(\text{Sentiment})$: $\{+: 8/12, -: 4/12\}$

IDX	Word1	Word2	Word3	Sentiment
1	4	2	0	+
2	7	3	1	+
3	6	4	3	+
4	5	2	2	+
5	8	5	3	+
6	4	7	5	+
7	9	7	2	+
8	12	2	3	+
9	3	3	7	-
10	0	4	8	-
11	4	3	15	-
12	1	4	10	-



Naïve Bayes: parameter estimation

Multinomial Distribution, example

Sub-dataset, select Sentiment=+



IDX	Word1	Word2	Word3	Sentiment
1	4	2	0	+
2	7	3	1	+
3	6	4	3	+
4	5	2	2	+
5	8	5	3	+
6	4	7	5	+
7	9	7	2	+
8	12	2	3	+

- Sum all values in rows → Frequency of words: {Word1: 55, Word2: 32, Word3: 19}
- Smooths Frequency with $\alpha = 5$: {Word1: 55+5, Word2: 32+5, Word3: 19+5}
- Smooths Frequency with $\alpha = 5$: {Word1: 60, Word2: 37, Word3: 24}
- $N_p = 60 + 37 + 24 = 121$
- $P(\text{words} | \text{Sentiment}=+)$: {Word1: 60/121, Word2: 37/121, Word3: 24/121}



Naïve Bayes: parameter estimation

Multinomial Distribution, example

Sub-dataset, select **Sentiment=+**



IDX	Word1	Word2	Word3	Sentiment
1	4	2	0	+
2	7	3	1	+
3	6	4	3	+
4	5	2	2	+
5	8	5	3	+
6	4	7	5	+
7	9	7	2	+
8	12	2	3	+

- Sum all values in rows → Frequency of words: {Word1: 55, Word2: 32, Word3: 19}
- Smooths Frequency with $\alpha = 5$: {Word1: 55+5, Word2: 32+5, Word3: 19+5}
- Smooths Frequency with $\alpha = 5$: {Word1: 60, Word2: 37, Word3: 24}
- $N_p = 60 + 37 + 24 = 121$
- $P(\text{words} | \text{Sentiment}=+)$: {Word1: 60/121, Word2: 37/121, Word3: 24/121}



Naïve Bayes: parameter estimation

Multinomial Distribution, example

Sub-dataset, select **Sentiment=-**



IDX	Word1	Word2	Word3	Sentiment
9	3	3	7	-
10	0	4	8	-
11	4	3	15	-
12	1	4	10	-

- Sum all values in rows → Frequency of words: {Word1: 8, Word2: 14, Word3: 40}
- Smooths Frequency with $\alpha = 5$: {Word1: 8+5, Word2: 14+5, Word3: 40+5}
- Smooths Frequency with $\alpha = 5$: {Word1: 13, Word2: 19, Word3: 45}
- $N_p = 13 + 19 + 45 = 77$
- $P(\text{words} | \text{Sentiment}=-)$: {Word1: 13/77, Word2: 19/77, Word3: 45/77}



Naïve Bayes: parameter estimation

Multinomial Distribution, example

Summary:

- $P(\text{Sentiment}): \{+: 8/12, -: 4/12\}$
- $P(\text{words}|\text{Sentiment}=+): \{\text{Word1}: 60/121, \text{Word2}: 37/121, \text{Word3}: 24/121\}$
- $P(\text{words}|\text{Sentiment}=-): \{\text{Word1}: 13/77, \text{Word2}: 19/77, \text{Word3}: 45/77\}$

Parameter estimation

$P(X_i|C)$: Gaussian distribution



Naïve Bayes: parameter estimation

Gaussian distribution

X_1	X_2	...	X_D	C
$x_1^{(1)}$	$x_2^{(1)}$	...	$x_D^{(1)}$	$c^{(1)}$
$x_1^{(2)}$	$x_2^{(2)}$	...	$x_D^{(2)}$	$c^{(2)}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$x_1^{(N)}$	$x_2^{(N)}$	...	$x_D^{(N)}$	$c^{(N)}$

(1) SELECT rows, given $C=c_k$ for each c_k

X_1	X_2	...	X_D	C
...	...	...	...	c_k
...	...	...	...	c_k
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
...	...	...	...	c_k

X_1	X_2	...	X_D	C
...	...	...	...	c_1
...	...	...	...	c_1
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
...	...	...	...	c_1

N_1

X_1	X_2	...	X_D	C
...	...	...	...	c_2
...	...	...	...	c_2
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
...	...	...	...	c_2

N_2

N_k

NOTE: $N_1 + N_2 + \dots + N_K = N$



Naïve Bayes: parameter estimation

Gaussian distribution

- C: Categorical Distribution
- $\text{Domain}(C) = \{c_1, c_2, \dots, c_K\}$
- $P(C): \{\frac{N_1}{N}, \frac{N_2}{N}, \dots, \frac{N_K}{N}\}$



Naïve Bayes: parameter estimation

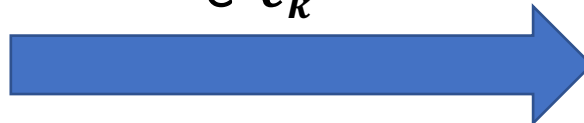
Gaussian distribution

Training dataset:

X_1	X_2	$\dots$	X_D	C
$x_1^{(1)}$	$x_2^{(1)}$	$\dots$	$x_D^{(1)}$	$c^{(1)}$
$x_1^{(2)}$	$x_2^{(2)}$	$\dots$	$x_D^{(2)}$	$c^{(2)}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$x_1^{(N)}$	$x_2^{(N)}$	$\dots$	$x_D^{(N)}$	$c^{(N)}$

(1) SELECT rows, given

$C=c_k$



X_1	X_2	$\dots$	X_D	C
$\dots$	$\dots$	$\dots$	$\dots$	c_k
$\dots$	$\dots$	$\dots$	$\dots$	c_k
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\dots$	$\dots$	$\dots$	$\dots$	c_k

NaiveBayes's assumption:

- X_i independent with X_j given C
- Define $\mathbf{x} = [X_1, X_2, \dots, X_D]$

- **Assumption:** $\mathbf{x} \sim N(\mathbf{x} | \boldsymbol{\mu}, \Sigma)$

- $$N(\mathbf{x} | \boldsymbol{\mu}, \Sigma) = \frac{1}{(2\pi|\Sigma|)^{\frac{1}{2}}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})\right\}$$



Naïve Bayes: parameter estimation

Gaussian distribution

- **Assumption:** $\mathbf{x} \sim N(\mathbf{x} | \boldsymbol{\mu}, \Sigma)$
- $N(\mathbf{x} | \boldsymbol{\mu}, \Sigma) = \frac{1}{(2\pi|\Sigma|)^{\frac{1}{2}}} \exp\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\}$
- Estimation:

$$\boldsymbol{\mu} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n$$

$$\Sigma = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu})(\mathbf{x}_n - \boldsymbol{\mu})^T$$

$\mathbf{x}_n$: samples in sub-datasets

Result of SELECTING rows, given $\mathbf{C}=\mathbf{c}_k$ for each $\mathbf{c}_k$

In numpy:

- $\boldsymbol{\mu}$: `numpy.mean`
- Σ : `numpy.cov`